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## Patent Public Search | Text View

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United States Patent Application Publication

20250281226

Kind Code

A1

Publication Date

September 11, 2025

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### **ELECTROSURGICAL SYSTEMS, DEVICES AND METHODS INCLUDING ECHOGENIC GUIDEWIRES**

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#### **Abstract**

An electrosurgical guidewire including a core wire. The core wire having a treated segment with an outer surface treated to impart an echogenic texture to the outer surface of the treated segment. The echogenic texture operating to increase ultrasound visibility of the treated segment.

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**Family ID:** 1000008533091

**Appl. No.:** 19/075881

**Filed:** March 11, 2025

#### **Related U.S. Application Data**

us-provisional-application US 63563667 20240311

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#### **Publication Classification**

**Int. Cl.:** A61B18/14 (20060101); A61B17/00 (20060101); A61B18/00 (20060101)

**U.S. Cl.:**

**CPC** A61B18/14 (20130101); A61B2017/00526 (20130101); A61B2018/00083 (20130101); A61B2018/00107 (20130101); A61B2018/00351 (20130101); A61B2018/00964 (20130101)

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#### **Background/Summary**

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/563,667, filed Mar. 11, 2024 (pending), the disclosure of which is incorporated by reference herein in its entirety.

## TECHNICAL FIELD

[0002] The present disclosure generally relates to guidewires and associated core wires useful in electrosurgical procedures, and devices, systems and methods including such guidewires and associated core wires.

## BACKGROUND

[0003] Guidewires and their associated core wires are often constructed with differing physical characteristics along portions or segments of their length. The more distal tip segment or portion may include any of a wide variety of features that help the guidewire meet performance requirements. The more proximal segment or portion may have a “coil over core” design, wherein a relatively thin coil wire is wrapped repeatedly circumferentially around a thicker and stiffer core wire that is generally straight but flexible to allow its manipulation through the patient's body, such as through vasculature. The core wire may be thinner (i.e., formed with a smaller outer diameter) at the distal tip segment or portion of the guidewire and thicker at the more proximal shaft segment to meet performance requirements. Purposes of the thin coil wire include keeping a constant outer guidewire diameter (despite possibly variable core wire diameters) and to facilitate guidewire flexibility and passage through tortuous anatomies. Coil wires can have a secondary, often unintended benefit of being highly echogenic. In this regard, the thin coil wire structure creates a surface texture that helps ultrasound waves emitted by an ultrasound transducer to “echo” or reflect back to the ultrasound transducer in a way that generates a useful image on a display viewed by the doctor and other medical professionals performing a procedure.

[0004] Some guidewires, at least for most of the lengths of their shaft segments, do not have a coil covering the core. This “core-only” guidewire design maximizes wire stiffness at a given guidewire diameter because none of the guidewire diameter is “lost” to flexible wraps of coil wire. Core-only guidewire designs are used to maximize guidewire stiffness within the constraints of a maximum outer diameter.

[0005] Electrosurgical guidewires are unique in that they require electrical insulation. Electrosurgical guidewires are not unique in that they may also need to be relatively stiff at small diameters. Core-only guidewire designs are typical for electrosurgical guidewire bodies because some of the “available” guidewire diameter is necessarily “lost” to the required insulation. Since adequate core wire insulation and stiffness are relatively more important, the coil wire may be eliminated along with its echogenicity.

[0006] Electrosurgical wire-based puncture of the interatrial septum is now commonly used for transseptal procedures, which may involve over-the-wire and/or next-to-the-wire delivery of devices to the left atrium. Next-to-the-wire delivery of devices to the left atrium is facilitated by echocardiographic visualization of the transseptal guidewire passing through the interatrial septostomy. In left atrial arrhythmia ablation procedures, for example, such visualization may be needed to facilitate passage of mapping and/or ablation catheters alongside the transseptal guidewire through the same interatrial septostomy. Similarly, when left atrial appendage occlusion procedures or mitral valve interventions are guided by intracardiac echocardiography (ICE), such visualization may be needed to facilitate passage of an ICE catheter alongside the transseptal guidewire through the same septostomy. If the transseptal guidewire is poorly echogenic, poor echocardiographic visualization of the transseptal guidewire causes difficulty advancing devices alongside the transseptal guidewire. If the procedure is a fluoroscopy-free one, the difficulty of advancing devices may be more significant, and there may be little recourse.

[0007] Currently available electrosurgical guidewires have core-only shaft segments that, within the constraints of a maximum outer diameter, optimize guidewire insulation and stiffness at the

expense of echogenicity. It would be beneficial to provide electrosurgical guidewires and associated devices, systems and methods that, within the constraints of a maximum outer diameter, provide improved visualization through echogenicity while still providing adequate electrical insulation and stiffness.

## SUMMARY

[0008] In one or more general aspects, electrosurgical guidewires are provided. One such guidewire includes a core wire having a segment with an outer surface treated to impart an echogenic texture to that segment. This makes that segment distinctly echogenic and therefore distinctly visible by an imaging system relative to an adjacent segment or segments of the wire. Particularly, the echogenic texture increases distinct ultrasound visibility of the treated segment of the wire.

[0009] The electrosurgical guidewire may have one or more optional and/or additional features. Several examples, which may provide one or more distinct benefits themselves or in any of various combinations are discussed herein. For instance, more than one segment of the outer surface may be treated to impart the echogenic texture. The outer surface may be treated without removal of a significant amount of core wire material and, therefore, without significant compromise of wire stiffness at the treated segment(s). The treatment methods may, for example, include one or more surface treatments such as bead blasting, dimpling, peening, grinding, laser treatment, sanding, or any other treatments or combinations of treatments that create echogenic irregularity of the wire. The outer surface may be covered by at least one of: spray coating, shrink tubing, or other electrical insulation materials. The wire may be used in transseptal procedures, and the treated segment may then further comprise a segment that lies across the interatrial septostomy after transseptal puncture. The length of the treated segment may be between about 1 cm and about 10 cm. The maximum outer diameter of the guidewire may be about 0.014 inches to 0.038 inches. The treated segment may further comprise circumferentially applied texturing treatment for creating echogenic irregularity of the outer surface that is oriented transverse to an ultrasound transducer. This orientation can further increase echogenicity and, therefore, optimize visualization. The electrosurgical guidewire may include a coiled wire wrapped around a portion of the core wire. In some embodiments, the coiled wire may be wrapped around a portion of the treated segment.

[0010] An electrosurgical system may include the electrosurgical guidewire and an apparatus for coupling the electrosurgical guidewire to an electrosurgical generator. The apparatus may include an elongated flexible conductive element, an activator unit for selectively controlling energy to the electrosurgical guidewire, and a coupler for removably coupling the electrosurgical guidewire to the apparatus. The electrosurgical system may include an electrosurgical generator.

[0011] In additional or alternative aspects, methods of using an electrosurgical guidewire are provided and generally comprise directing an electrosurgical guidewire into a patient. The electrosurgical guidewire includes a core wire having a treated segment with an outer surface treated to impart an echogenic texture thereto, such as in any of the broader and/or specific manners discussed herein. The echogenic texture increases ultrasound visibility of the treated segment. The method includes visualizing the treated segment using an ultrasonic imaging system.

[0012] The methods may include, for example, one or more of the optional features discussed herein. Illustrative examples are discussed below, but of course other features or options may be included instead of or in addition to those specifically discussed herein. More than one segment of the outer surface may be treated to impart the echogenic texture, and visualizing the segment may further comprise visualizing more than one segment. The outer surface may be treated without removal of a significant amount of wire material and without significant compromise of wire stiffness. The outer surface may be covered by at least one of: spray coating, shrink tubing, or other electrical insulation materials. The outer surface may be surface treated by at least one of: bead blasting, dimpling, peening, grinding, laser treatment, sanding, or any of these and/or other treatments or combinations of treatments that create echogenic irregularity of the wire.

[0013] The method may further comprise a transseptal procedure, and in this case may further comprise visualizing the segment lying across the interatrial septostomy after transseptal puncture. The length of the segment may be between about 1 cm and about 10 cm. The maximum outer diameter of the guidewire may be about 0.014 inches to 0.038 inches. The treated segment may further comprise circumferentially applied texturing treatment for creating echogenic irregularity of the outer surface wherein the surface treatment is oriented transverse to an ultrasound transducer, and visualizing the segment further comprises visualizing the transversely oriented echogenic irregularity. In some embodiments, the electrosurgical guidewire may include a coiled wire wrapped around a portion of the core wire. The method may include visualizing the coiled wire. The coiled wire may be wrapped around a portion of the treated segment.

[0014] In some embodiments of the method, the electrosurgical guidewire may be part of an electrosurgical system. The electrosurgical system may include the electrosurgical guidewire and an apparatus for coupling the electrosurgical guidewire to an electrosurgical generator. The apparatus may include an elongated flexible conductive element, an activator unit for selectively controlling energy to the electrosurgical guidewire, and a coupler for removably coupling the electrosurgical guidewire to the apparatus. The electrosurgical system may also include an electrosurgical generator.

[0015] Additional and/or alternative aspects of the invention may include systems that include guidewires within the scope of this invention. As an example, such a system may include a radiofrequency (RF) energy generation unit useful in supplying RF energy to the guidewire so that the tip can easily penetrate tissue such as septal tissue of the heart. Additional system components may include visualization equipment/devices used during such procedures to provide one or more images of the wire segment(s) during a medical procedure.

[0016] Various additional features and advantages will become readily apparent to those of ordinary skill in the art upon review of the following detailed description of the illustrative embodiments. Any of the features and functions described herein may be applied to any of the disclosed embodiments or methods.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 shows an illustrative electrosurgical transseptal puncture guidewire.

[0018] FIG. 1A is a perspective view of an illustrative system for delivering RF energy to tissue during a medical procedure including the electrosurgical guidewire of FIG. 1.

[0019] FIG. 2 is a cross sectional view taken along line 2-2 of FIG. 1.

[0020] FIG. 3 shows an alternative illustrative electrosurgical transseptal puncture guidewire.

[0021] FIG. 4 illustrates a heart model with an electrosurgical transseptal puncture guidewire and an intracardiac echocardiography catheter inserted into the model.

[0022] FIG. 5A is an intracardiac echocardiography image of an electrosurgical transseptal puncture guidewire with an untreated guidewire core.

[0023] FIG. 5B is an intracardiac echocardiography image of an electrosurgical transseptal puncture guidewire with a segment treated to create an echogenic texture operating to increase the ultrasound visibility of the treated segment.

### DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

[0024] FIG. 1 shows an illustrative electrosurgical transseptal puncture guidewire **100** and FIG. 2 illustrates a cross section of the electrosurgical guidewire **100**. The electrosurgical guidewire **100** includes a core wire **102** with an outer surface **104** covered by electrical insulation **106**. In an illustrative aspect, the electrosurgical guidewire **100** may be of any desired design in terms of being, for example, “core-only” or “coil-over-core.” For convenience only, this detailed description

uses the term “core,” but the disclosure herein is equally applicable, as will become apparent to those of ordinary skill, to “core-only” segments of coil-over core guidewire designs.

[0025] The electrosurgical guidewire **100** includes a distal wire segment **110**, a more proximal wire segment **120**, and a shaft segment **130**. The distal wire segment **110** includes a bare or exposed distal tip **112** or terminus of the electrosurgical guidewire **100** serving as the “active electrode” in the system, where electric current density is highly concentrated to effect tissue vaporization. The electrosurgical guidewire **100** depicted in FIG. **1** includes a pigtail shaped distal wire segment **110**, but a J-tipped, straight-tipped, or other design configuration may be used. The shaft segment **130** has a bare or exposed portion **132** for purposes of electrical connection to, for example, an electrosurgical generator unit supplying radiofrequency (RF) energy. The electrosurgical guidewire **100** together with an electrosurgical generator unit, along with any other desired components useful during medical procedures involving the use of the electrosurgical guidewire **100** and electrosurgical generator unit, can comprise a system in accordance with various embodiments. In some embodiments, these bare or exposed portions **112**, **132** of the core wire **102** may be located at respective distal and proximal terminus points or ends. However, in other embodiments, for example, one or both of these bare or exposed portions **112**, **132** may be located near but not at a terminus point. In some embodiments, a bare or exposed portion, serving as an active electrode, may be located remotely from the terminus points or ends. The remainder of the electrosurgical guidewire **100** must be insulated to minimize charge dispersion and related loss of electrosurgical effect.

[0026] In this illustrative embodiment, the outer surface **104** of the core wire **102** of the more proximal wire segment **120** is treated to impart an echogenic texture to the more proximal wire segment **120**. The echogenic texture increases the visibility by an imaging device such as an ultrasound-based imaging device. The treatment of the surface **104** of the core wire **102** may be usefully applied to any core-only segment(s) of the electrosurgical guidewire **100**, including more distal wire segments **110**, more proximal wire segments **120**, and/or shaft segments **130**. The outer diameter of the core wire **102** is an important determinant of mechanical guidewire performance (e.g. mechanical characteristics such as strength and flexibility). The treatment of the surface **104** of the core wire **102** may be completed without removal of a significant amount of core wire material and without significant compromise of core wire stiffness. Options for segmental treatment of the surface **104** of the core wire **102** may include bead blasting, dimpling, peening, grinding, laser treatment, sanding, and any other treatments or combinations of treatments that create echogenic irregularity in the otherwise smooth surface of the core wire **102**. The insulation **106** is the main determinant of electrosurgical guidewire performance. Treated segments of the core wire **102** may or may not be covered by spray coating, shrink tubing, or other electrical insulation materials.

[0027] FIG. **1A** is a perspective view of an illustrative system for delivering RF energy to tissue during a medical procedure. Further illustrative details of this exemplary system may be found in U.S. patent application Ser. No. 18/243,927, filed on Sep. 8, 2023, the disclosure of which is hereby incorporated by reference herein. In this illustrative embodiment, the electrosurgical guidewire **100**, an electrosurgical unit **200**, and an apparatus **202** including an elongated flexible conductive element **204**, an activator unit **206**, and a coupler **208** form the system **210** which is configured for performing a medical procedure. The elongated flexible conductive element **204** is an electrically insulated conductor, such as a wire, configured for transmitting electricity or RF energy. In some embodiments, the elongated flexible conductive element **204** may be a cable including an insulated wire or wires and having a protective casing. The elongated flexible conductive element **204** includes a proximal end **212** including an electrosurgical unit connector **214**, and a distal end **216** coupled to the coupler **208**. In this illustrative embodiment, the coupler **208** is configured to removably couple the electrosurgical guidewire **100** to the elongated flexible conductive element **204**. The electrosurgical unit connector **214**, capable of attaching to

conventional RF energy generating units for delivering RF energy, may releasably connect to the electrosurgical unit **200**. Many commercially available electrosurgical units include a standardized receptacle, such as a monopolar accessory receptacle. The electrosurgical unit connector **214** may be configured to couple to any one of a plurality of electrosurgical units with standardized receptacles. Therefore, the subsystem or assembly comprising, for example, the electrosurgical guidewire **100**, and the apparatus **202** including the elongated flexible conductive element **204**, activator unit **206**, and coupler **208** may be physically coupled to one of several different standardized receptacles of an electrosurgical RF generating unit. In the illustrative embodiment the elongated flexible conductive element **204** has a mid-portion **218** that is connected to the activator unit **206**. The activator unit **206** is situated at a location spatially separated from the proximal end **212** of the elongated flexible conductive element **204** and, therefore also spatially separated from the electrosurgical unit **200**, by a segment of the elongated flexible conductive element **204**. The activator unit **206** also may be spatially separated from the coupler **208** by another segment of the elongated flexible element **204**, as shown in FIG. **1A** or, optionally, the activator unit **206** may be integrated with or otherwise fixed to the coupler **208**. In some embodiments, the activator unit **206** may be located at the proximal end **212** of the elongated flexible conductive element **204** adjacent to or otherwise fixed to the electrosurgical unit connector **214**. The activator unit **206** includes a switch element **220**, such as a push button or other element, allowing the user to selectively activate the flow of RF energy. The apparatus **202** includes an electroanatomical mapping connector **222** that is capable of connecting to electroanatomical mapping systems.

[0028] FIG. **3** shows an alternative illustrative electrosurgical transseptal puncture guidewire **100'**. Generally, the electrosurgical guidewire **100'** is similar in construction and operation to the electrosurgical guidewire **100** described above and the electrosurgical guidewire **100'** may be substituted for other electrosurgical guidewires, or any electrosurgical guidewire **100'** may be used, in various other exemplary embodiments according to the present disclosure. Like reference numbers refer to like components. For brevity, the following description minimizes redundant description and focuses on the differences between the electrosurgical guidewire **100** and the electrosurgical guidewire **100'**.

[0029] The electrosurgical guidewire **100'** includes a core wire **102'** with an outer surface **104'** covered by electrical insulation **106'**. The electrosurgical guidewire **100'** includes a distal wire segment **110'**, a more proximal wire segment **120'**, and a shaft segment **130'**. The distal wire segment **110'** includes a bare or exposed distal tip **112'** or terminus of the electrosurgical guidewire **100'** serving as the “active electrode” in the system. The electrosurgical guidewire **100'** depicted in FIG. **3** includes a straight distal wire segment **110'**, but a pigtail shaped, J-tipped, or other design configuration may be used.

[0030] In this illustrative embodiment, distal wire segment **110'** includes a coiled wire **114'** located over the core wire **102'**. At this position, the coiled wire **114'** imparts an echogenic texture to the portion of the distal wire segment **110'** covered by the coiled wire **114'**. The coiled wire **114'** increases the ultrasound visibility of the portion of the distal wire segment **110'** covered by the coiled wire **114'**. For example, like the electrosurgical guidewire **100**, this coiled wire segment may provide a distinct segment of increased echogenicity relative to one or more adjacent portions of the electrosurgical guidewire **100'**. This may provide a targeted visualization of that enhanced echogenic segment relative to the one or more adjacent segments. Any portion of the distal wire segment **110'**, including the portion covered by the coiled wire **114'**, may be treated to impart an echogenic texture.

[0031] The outer surface of the core wire of the more proximal wire segment **120'** is treated to impart an echogenic texture to the more proximal wire segment **120'**. The treatment of the surface of the core wire may be usefully applied to any core-only segment(s) of the electrosurgical guidewire **100'**, including more distal wire segments **110'**, more proximal wire segments **120'**,

and/or shaft segments **130'**.

[0032] FIG. **4** illustrates a water-filled heart model **10** useful in benchtop assessment of the echogenicity (and echocardiographic visibility) of transseptal guidewires. Illustrated is an introducer set **60** extending from the inferior vena cava **12** into the right atrium **14** with the electrosurgical guidewire **100** extending from the tip of a dilator **62** of the introducer set **60**, through the interatrial septum **16**, into the left atrium **18**, and, in this case, into the left atrial appendage **20**. Also shown is an ICE catheter **70** extending from the inferior vena cava **12** into the right atrium **14**. For electrosurgical transseptal puncture guidewires that are used in transseptal procedures, the segment **140** of the electrosurgical guidewire **100** that lies across the interatrial septum **16** after transseptal puncture is the most clinically relevant segment.

[0033] Depending on the size of the left atrium **18** (which determines how much wire might be advanced into the left atrium **18** after transseptal puncture), the relevant segment **140** may ideally be about 6 cm long, and a preferred but exemplary range of the relevant segment length is about 1 cm to about 10 cm. For electrosurgical transseptal puncture guidewires **100** used in transseptal procedures, the maximum outer diameter of the guidewire **100** may be constrained to less than about 0.014 to 0.038 inches to maintain compatibility with available introducer sets.

[0034] Since the long axes of transseptal guidewires and right atrial ICE catheters are roughly parallel in the heart when both devices are inserted from a femoral vein, circumferentially applied texturing treatment (analogous to wraps of coil wire around a core wire) may create echogenic irregularity of the core wire's surface that is generally orthogonal to the ICE catheter's ultrasound transducer. Other transverse orientations may be used instead. The circumferentially applied, transversely oriented texturing treatment may increase echogenicity (and echocardiographic visibility) more substantially than a longitudinally applied, or less transversely oriented texturing treatment.

[0035] A water-filled heart model **10**, illustrated in FIG. **4**, was used to obtain the intracardiac echocardiography images in FIGS. **5A** and **5B**. The intracardiac echocardiography image in FIG. **5A** was obtained using an untreated guidewire **300** having a core wire with a relatively smooth surface. The intracardiac echocardiography image in FIG. **5B** was obtained using an illustrative embodiment of an electrosurgical transseptal puncture guidewire with a segment **120** of the core wire treated to create an echogenic texture operating to increase the ultrasound visibility of the treated segment. The guidewire in FIG. **5B** is much more echodense (easier to see) than the guidewire **300** in FIG. **5A**. The echogenic texture on the illustrative guidewire embodiment used in FIG. **5B** was created by circumferentially "sanding" the guidewire core by using a drill motor to spin it between opposed pieces of 80-grit sandpaper. The sanding process imparted a palpable texture to the wire's surface but did not remove a significant volume of material from the wire. The circumferentially sanded guidewire core depicted in FIG. **5B** is substantially more echogenic (and echocardiographically visible) than the untreated guidewire core depicted in FIG. **5A**. It should be understood that this is just one illustrative example of a treatment applied to a core wire to create an echogenic texture and other treatments may be applied to a core wire to generate the disclosed echogenic texture. It should be noted that the images of FIGS. **5A** and **5B** include reverberation artifacts **310** which is a type of ultrasound imaging artifact that occurs when sound waves bounce back and forth between two highly reflective surfaces, creating multiple echoes. These reverberation artifacts **310** do not represent any structure of the heart model or guidewires.

[0036] While the present invention has been illustrated by the description of specific embodiments thereof, and while the embodiments have been described in considerable detail, it is not intended to restrict or in any way limit the scope of the appended claims to such detail. The various features discussed herein may be used alone or in any combination within and between the various embodiments. Additional advantages and modifications will readily appear to those skilled in the art. The invention in its broader aspects is therefore not limited to the specific details, representative apparatus and methods and illustrative examples shown and described. Accordingly,

departures may be made from such details without departing from the scope or spirit of the general inventive concept.

## Claims

1. An electrosurgical guidewire, comprising: a core wire having a treated segment with an outer surface treated to impart an echogenic texture thereto, the echogenic texture operating to increase ultrasound visibility of the treated segment.
2. The electrosurgical guidewire of claim 1, wherein the core wire comprises more than one treated segment.
3. The electrosurgical guidewire of claim 1, wherein the outer surface of the treated segment is treated without removal of a significant amount of wire material and without significant compromise of wire stiffness.
4. The electrosurgical guidewire of claim 1, further comprising a coating of at least one of: spray coating, shrink tubing, or other electrical insulation materials.
5. The electrosurgical guidewire of claim 1, wherein the outer surface of the treated segment is treated by at least one of: bead blasting, dimpling, peening, grinding, laser treatment, sanding, or any other treatments or combinations of treatments that create echogenic irregularity of the treated segment.
6. The electrosurgical guidewire of claim 1, wherein the core wire is used in transseptal procedures, and a portion of the treated segment lies across the interatrial septostomy after transseptal puncture.
7. The electrosurgical guidewire of claim 6, wherein the length of the treated segment is between about 1 cm and about 10 cm.
8. The electrosurgical guidewire of claim 1, wherein the maximum outer diameter of the guidewire is about 0.014 inches to 0.038 inches.
9. The electrosurgical guidewire of claim 1, wherein the treated segment further comprises circumferentially applied texturing treatment for creating echogenic irregularity of the outer surface that is oriented transverse to an ultrasound transducer.
10. The electrosurgical guidewire of claim 1, further comprising a coiled wire wrapped around a portion of the core wire.
11. The electrosurgical guidewire of claim 10, wherein the coiled wire is wrapped around a portion of the treated segment.
12. An electrosurgical system comprising: the electrosurgical guidewire of claim 1; and an apparatus for coupling the electrosurgical guidewire to an electrosurgical generator, the apparatus including: an elongated flexible conductive element, an activator unit for selectively controlling energy to the electrosurgical guidewire, and a coupler for removably coupling the electrosurgical guidewire to the apparatus.
13. The electrosurgical system of claim 12, further comprising an electrosurgical generator.
14. A method of using an electrosurgical guidewire, comprising: directing an electrosurgical guidewire into a patient, the electrosurgical guidewire including a core wire having a treated segment with an outer surface treated to impart an echogenic texture thereto, the echogenic texture operating to increase ultrasound visibility of the treated segment, and visualizing the treated segment using an ultrasonic imaging system.
15. The method of claim 14, wherein the core wire includes more than one treated segment, and visualizing the treated segment further comprises visualizing more than one treated segment.
16. The method of claim 14, wherein the outer surface of the treated segment is treated without removal of a significant amount of wire material and without significant compromise of wire stiffness.
17. The method of claim 14, wherein the electrosurgical guidewire includes a coating of at least one of: spray coating, shrink tubing, or other electrical insulation materials.



- 18.** The method of claim 14, wherein the outer surface of the treated segment is treated by at least one of: bead blasting, dimpling, peening, grinding, laser treatment, sanding, or any other treatments or combinations of treatments that create echogenic irregularity of the treated segment.
- 19.** The method of claim 14, wherein the method further comprises a transseptal procedure, and visualizing the treated segment further comprises visualizing the treated segment lying across the interatrial septostomy after transseptal puncture.
- 20.** The method of claim 19, wherein the length of the treated segment is between about 1 cm and about 10 cm.
- 21.** The method of claim 14, wherein the maximum outer diameter of the guidewire is about 0.014 inches to 0.038 inches.
- 22.** The method of claim 14, wherein the treated segment further comprises circumferentially applied texturing treatment for creating echogenic irregularity of the outer surface that is oriented transverse to an ultrasound transducer, and visualizing the treated segment further comprises visualizing the transversely oriented echogenic irregularity.
- 23.** The method of claim 14, wherein the electrosurgical guidewire further comprises a coiled wire wrapped around a portion of the core wire, and the method further comprises visualizing the coiled wire.
- 24.** The method of claim 23, wherein the coiled wire is wrapped around a portion of the treated segment.
- 25.** The method of claim 14, wherein the electrosurgical guidewire is part of an electrosurgical system comprising the electrosurgical guidewire and an apparatus for coupling the electrosurgical guidewire to an electrosurgical generator, the apparatus includes an elongated flexible conductive element, an activator unit for selectively controlling energy to the electrosurgical guidewire, and a coupler for removably coupling the electrosurgical guidewire to the apparatus.
- 26.** The method of claim 25, wherein the electrosurgical system further comprises an electrosurgical generator.
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